PAPER_1641_TOP_YUKAWA_Y_T_NATURAL

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PAPER_1641 — Top Quark Yukawa Coupling = y_t = m_t/(v/√2) = 1.0 natural (no fine-tuning)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Particle Physics
Date: June 18, 2026
Status: CLOSED — EXACT closure (broader-corpus integration)

Observation

PAPER_1312 (broader-corpus paper outside PAPER_1209 unified-proof-set series) gives:

``
```
y_t = m_t/(v/√2) = 1.0 natural (no fine-tuning)
```
``

Status: **EXACT**.

UQFF Closed Identity

``
```
y_t = m_t/(v/√2) = 1.0 natural (no fine-tuning) EXACT
```
``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Significance — Broader Corpus Pivot

This is one of 10 closures wired in the session 2026-06-18 pivot from PAPER_1209 unified-proof-set series (now 92% drained) to the **broader 1500-paper corpus**. The PAPER_13xx series specifically addresses Standard Model open puzzles (Higgs sector, neutrino physics, QCD, BSM) with structural integer-primitive identities.

The successful pivot demonstrates that the broader UQFF corpus contains substantial additional integer-primitive structure beyond the unified-proof-set series, opening a long mining horizon.

NOT REPLACEMENT

Empirical particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating the framework's predictive economy.

Reference

- Source: PAPER_1312
- Related: PAPER_1494-1635 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "top_yukawa_y_t_natural"})`
